Question 1: Web Server CPU Utilization and Request Processing Capacity

A company runs a high-traffic e-commerce website with a web server that has **16 CPU cores**, each operating at **3.2 GHz**. The server processes an average of **350 HTTP requests per second**, with each request consuming **5 milliseconds of CPU time** on a single core. Due to increased traffic, the company anticipates a **40% rise** in incoming requests.

The IT auditor must determine whether the server can handle this projected increase without exceeding **80% CPU utilization**. If the projected load surpasses this threshold, a new server must be added. Should the company invest in an additional server?

Solution:

1. **Calculate Current CPU Usage:**
   - **CPU Time Required per Request:** 5 ms/request
   - **Total CPU Time per Second:** 350 requests/sec × 5 ms/request = **1,750 ms/sec**
   - **Total Available CPU Time per Second:** 16 cores × 1,000 ms/sec = **16,000 ms/sec**
   - **Current CPU Utilization:** (1,750 / 16,000) × 100 = **10.94%**
2. **Projected CPU Usage with 40% Traffic Increase:**
   - **New Requests per Second:** 350 × 1.40 = **490 requests/sec**
   - **New Total CPU Time Required:** 490 × 5 ms = **2,450 ms/sec**
   - **New CPU Utilization:** (2,450 / 16,000) × 100 = **15.31%**

Answer:

Since the projected CPU utilization remains well below the **80% threshold**, the company **does not** need an additional server at this time.

Question 2: Web Server Response Time and Performance Bottlenecks

A web server has been experiencing slow response times during peak hours. The server is designed to handle **10,000 concurrent connections**, and the average response time per request is **0.25 seconds**. However, during peak traffic, logs show that **30% of the requests take 1.2 seconds instead of 0.25 seconds**.

An IT auditor needs to determine how this increased response time impacts the overall **average response time** of the server. If the acceptable response time is **0.5 seconds**, does the server meet the performance criteria?

Solution:

1. **Calculate Weighted Average Response Time:**
   - **70% of requests process in 0.25 sec:** 0.70 × 0.25 = **0.175 sec**
   - **30% of requests process in 1.2 sec:** 0.30 × 1.2 = **0.36 sec**
   - **Total Average Response Time:** 0.175 + 0.36 = **0.535 sec**

Answer:

Since the **new average response time (0.535 sec) exceeds the acceptable threshold of 0.5 seconds**, the server **does not meet the performance criteria** and needs optimization, possibly through load balancing or hardware upgrades.

Question 3: Web Server Data Transfer Costs in a Cloud Environment

A company hosts its web server on a cloud provider that charges **$0.08 per GB** for outbound data transfer. The server transfers an average of **120 TB per month**, but the company expects a **25% increase** in data traffic due to a new marketing campaign. The IT auditor must calculate the **new estimated monthly cost** and determine whether it fits within the allocated budget of **$12,000 per month** for data transfer.

Solution:

1. **Current Monthly Cost:**
   - **Convert TB to GB:** 120 TB × 1,000 GB/TB = **120,000 GB**
   - **Cost for Current Usage:** 120,000 × $0.08 = **$9,600**
2. **Projected Data Usage After 25% Increase:**

- **New Data Transfer:** 120,000 GB × 1.25 = **150,000 GB**
- **New Monthly Cost:** 150,000 × $0.08 = **$12,000**

Answer:
The **new estimated monthly cost ($12,000) reaches the budget limit**. Any additional increase in traffic would result in overspending, so cost-saving strategies like content delivery networks (CDNs) or data compression should be considered.

Question 4: Web Server Power Consumption and Energy Cost Analysis

A data center operates **50 web servers**, each consuming **650 watts per hour**. The facility runs 24/7, and electricity costs **$0.14 per kWh**. Due to sustainability initiatives, the company plans to **replace these servers with energy-efficient models** consuming **30% less power**.
An IT auditor must calculate the **current monthly power cost**, the **expected savings**, and determine if the investment in new servers (costing $80,000) is justified if the payback period is within **2 years**.
Solution:

1. **Current Monthly Power Consumption:**
   - **Power per Server per Day:** 650 W × 24 hours = **15.6 kWh**
   - **Total for 50 Servers:** 15.6 × 50 = **780 kWh/day**
   - **Monthly Consumption:** 780 × 30 = **23,400 kWh**
   - **Monthly Cost:** 23,400 × $0.14 = **$3,276**
2. **Projected Monthly Cost After Replacing Servers:**
   - **New Power Usage:** 650 W × (1 - 0.30) = **455 W per server**
   - **New Monthly Consumption:** (455 × 24 × 50 × 30) ÷ 1,000 = **16,380 kWh**
   - **New Monthly Cost:** 16,380 × $0.14 = **$2,293.20**
3. **Monthly Savings:**
   - $3,276 - $2,293.20 = **$982.80**
   - **Annual Savings:** $982.80 × 12 = **$11,793.60**
4. **Payback Period:**
   - Investment Cost: $80,000
   - Payback Period = $80,000 ÷ $11,793.60 ≈ **6.8 years**

Answer:
Since the **payback period (6.8 years) exceeds 2 years**, the investment **is not justified** unless further cost reductions or rebates are available.

Question 5: Web Server Disk Space Utilization for Website Data

A web server hosts **450 websites**, each consuming an average of **2.4 GB** of storage. The total storage available on the server is **2 PB (petabytes)**. The hosting company expects an **annual growth rate of 15%** in storage usage.
The IT auditor must determine how many years it will take before **50% of the server's total capacity** is consumed.
Solution:

1. **Current Storage Usage:**
   - 450 websites × 2.4 GB = **1,080 GB = 1.08 TB**
2. **Capacity Limit (50% of 2 PB):**
   - 50% × 2 PB = **1 PB (1,000 TB)**
3. **Using the Compound Growth Formula:**
   - **A = P(1 + r)^t**, where:
     - A = 1,000 TB (half capacity)
     - P = 1.08 TB (current usage)
     - r = 15% (0.15)

- Solve for t:
  - $1{,}000 = 1.08 \times (1.15)^t$
  - $(1{,}000 / 1.08) = (1.15)^t$
  - $\log(925.93) / \log(1.15) \approx$ **20.58 years**

Answer:
The server will reach **50% capacity in approximately 20.6 years**, meaning upgrades will not be necessary for a long time.

Question 6: Estimating Web Server Response Time Under Load

A company's web server currently handles an average of **10,000 requests per second** with an average response time of **250 milliseconds (ms)** per request. Due to an upcoming marketing campaign, the website is expected to experience a **60% increase in traffic**. The IT team wants to estimate the new average response time if the server's total processing capacity remains unchanged.
Web server response time typically increases non-linearly with higher traffic due to increased CPU and memory utilization. A standard approximation is that response time grows proportionally to the increase in traffic squared. Given this, the team must determine whether the current infrastructure can handle the expected load efficiently or if scaling is necessary.

Solution:

1. **Define the Traffic Increase:**
   - Current requests per second (R_0) = **10,000 RPS**
   - Expected increase = **60%** → New requests per second (R_1) = **$10{,}000 \times 1.6 = 16{,}000$ RPS**
2. **Calculate Expected Response Time Increase:**
   - Response time typically increases as the square of the traffic increase:
   - New response time (T_1) = $\mathbf{T_0 \times (R_1 / R_0)^2}$
   - $T_1 = \mathbf{250\ ms \times (16{,}000 / 10{,}000)^2}$
   - $T_1 = \mathbf{250\ ms \times (1.6)^2 = 250 \times 2.56}$
   - $T_1 = \mathbf{640\ ms}$

Answer:
With a 60% traffic increase, the estimated new average response time will be **640 milliseconds per request**, more than double the previous time. The IT team should consider optimizing or upgrading the server infrastructure.

Question 7: Web Server Load Balancing Strategy

A company operates **four web servers** behind a load balancer, each capable of handling a maximum of **30,000 requests per minute**. During peak hours, the company's website receives **100,000 requests per minute**. If one web server fails, the IT team must determine whether the remaining servers can handle the traffic or if additional resources are required.
To prevent service degradation, the load balancer must distribute traffic evenly among the available servers. If a single server failure would exceed the capacity of the remaining servers, the company must either add redundancy or enable auto-scaling.

Solution:

1. **Calculate Current Load Distribution:**
   - Total capacity with all 4 servers = **$30{,}000 \times 4 = 120{,}000$ requests per minute**
   - Incoming traffic = **100,000 requests per minute**
   - Load per server with all working: **$100{,}000 \div 4 = 25{,}000$ requests per minute per server**
2. **Calculate Load Per Server After One Failure:**
   - Remaining servers = **3**
   - New load per server = **$100{,}000 \div 3 = 33{,}333$ requests per minute per server**

3. **Compare Against Server Capacity:**
    - Each server can handle a maximum of **30,000 requests per minute**
    - Required capacity per server (33,333) **exceeds** the limit (30,000)

Answer:
If one server fails, the remaining three servers **cannot** handle the traffic as they would need to process **33,333 requests per minute each**, exceeding their 30,000 limit. The company must add an additional server or enable auto-scaling to prevent downtime.

Question 8: Data Transfer Cost for Web Server Traffic

A company hosts a web application that serves **1 million page views per day**. Each page view requires downloading an average of **3 MB** of data, including images, scripts, and HTML content. The company's hosting provider charges **$0.09 per GB** for outbound data transfer. The IT team wants to estimate the monthly cost of data transfer.
Since web traffic volume fluctuates, the IT team assumes the daily traffic estimate is stable throughout the month. If costs exceed their budget, they may consider using a content delivery network (CDN) to reduce bandwidth expenses.

Solution:

1. **Calculate Daily Data Transfer:**
    - 1,000,000 page views/day × 3 MB per page = **3,000,000 MB/day**
    - Convert to GB: **3,000,000 ÷ 1,024 = 2,929.69 GB/day**
2. **Calculate Monthly Data Transfer:**
    - **2,929.69 GB/day × 30 days = 87,890.63 GB/month**
3. **Calculate Monthly Cost:**
    - **87,890.63 GB × $0.09/GB = $7,910.16**

Answer:
The estimated monthly data transfer cost is **$7,910.16**. The company should consider a CDN or data compression to reduce bandwidth costs.

Question 9: Server CPU Utilization During Peak Load

A web server is configured with a **16-core CPU**, where each core can handle **1,500 requests per second** before reaching full utilization. During peak hours, the web application experiences **20,000 requests per second**. The IT team wants to determine whether the current server configuration is sufficient or if additional resources are needed.
Overloading the CPU could lead to slow response times, dropped requests, or complete server failure. The IT team must ensure that peak traffic remains within the server's processing capability.

Solution:

1. **Calculate Total Processing Capacity:**
    - 16 cores × 1,500 requests per second = **24,000 RPS maximum capacity**
2. **Compare with Peak Traffic:**
    - Peak traffic = **20,000 RPS**
    - 20,000 RPS is **83.33% of total capacity** (20,000 ÷ 24,000 × 100)

Answer:
At peak load, CPU utilization reaches **83.33%**, which is within safe limits but leaves little room for growth. The IT team should monitor performance and consider adding more CPU resources for future scalability.

Question 10: Web Server Memory Consumption Calculation

A company operates a web server with **64 GB of RAM**. The server hosts a high-traffic website where each active session consumes an average of **2 MB** of memory. The IT team wants to determine the maximum number of concurrent sessions the server can handle before running out of memory.

Insufficient memory can cause performance issues, including increased swap usage, slower response times, or crashes. If the number of concurrent users approaches the memory limit, upgrading RAM or optimizing session management may be required.

Solution:

1. **Convert Total RAM to MB:**
   - 64 GB × 1,024 MB/GB = **65,536 MB**
2. **Calculate Maximum Concurrent Sessions:**
   - **65,536 MB ÷ 2 MB per session = 32,768 sessions**

Answer:

The web server can support **32,768 concurrent sessions** before exhausting its memory. If traffic is expected to grow, the company should consider increasing RAM or optimizing memory usage.

Final Thoughts

These questions cover critical aspects of web server auditing, including **performance scaling, load balancing, data transfer costs, CPU utilization, and memory constraints**. These calculations help IT teams make **informed decisions** about infrastructure upgrades and optimizations. Let me know if you need further refinements! 🚀